DARWIN AND EVOLUTION

Module map

- Module 12: Darwin and Evolution
- Connected lab: lab-12_darwin-evolution.md
- Use this deck as the visual path through the module's evidence and retrieval work.

Module 12

Darwin and Evolution

1. Historical explanation for adaptation
2. Variation and heritability
3. Natural selection mechanism
4. Evidence for common ancestry
5. Fitness as reproductive success

Lab evidence

lab-12_darwin-evolution.md

DARWIN AND EVOLUTION

Learning objectives

- Explain natural selection using variation, heritability, and reproductive success.
- Define fitness in evolutionary terms.
- Distinguish individual change from population evolution.
- Use multiple evidence types to support common ancestry.
- Apply natural selection to a concrete trait example.

OBJECTIVE 1
Explain natural selection using variation, heritability, and reproductive success.

OBJECTIVE 2
Define fitness in evolutionary terms.

OBJECTIVE 3
Distinguish individual change from population evolution.

OBJECTIVE 4
Use multiple evidence types to support common ancestry.

OBJECTIVE 5
Apply natural selection to a concrete trait example.

DARWIN AND EVOLUTION

Topic sequence

- Historical explanation for adaptation: Darwinian evolution explains adaptation through descent with modification.
- Variation and heritability: Natural selection requires variation, heritability, and unequal reproductive success.
- Natural selection mechanism: Fitness means relative reproductive success in a particular environment.
- Evidence for common ancestry: Fossils, homology, biogeography, and molecular evidence support common ancestry.
- Fitness as reproductive success: Evolution changes populations over generations, not individual organisms during life.

01
Historical explanation for adaptation
Darwinian evolution explains adaptation through descent with modification.

02
Variation and heritability
Natural selection requires variation, heritability, and unequal reproductive success.

03
Natural selection mechanism
Fitness means relative reproductive success in a particular environment.

04
Evidence for common ancestry
Fossils, homology, biogeography, and molecular evidence support common ancestry.

05
Fitness as reproductive success
Evolution changes populations over generations, not individual organisms during life.

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Module 12: Darwin and Evolution Evidence Map

- Module focus: Darwin and Evolution
- Central claim: Darwin and Evolution explains natural selection from variation, heritability, fitness, and ancestry evidence.
- Key nodes to connect: Historical explanation for adaptation, Variation and heritability, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-12_darwin-evolution.md supplies an evidence surface for the map.

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Module 12: Darwin and Evolution Reasoning Sequence

- Module focus: Darwin and Evolution
- Trace the model: Historical explanation for adaptation -> Variation and heritability -> Natural selection mechanism -> Evidence for common ancestry
- Inputs become outputs: Historical explanation for adaptation, Variation and heritability -> Explain natural selection using variation, heritability, and reproductive success., Define fitness in evolutionary terms.
- Feedback check: lab-12_darwin-evolution.md evidence checks whether students can explain natural selection mechanism.

DARWIN AND EVOLUTION

Terms as evidence handles

- Evolution: Change in heritable traits of populations across generations.
- Natural selection: Nonrandom differences in survival or reproduction tied to heritable traits.
- Adaptation: Heritable trait shaped by selection in a context.
- Fitness: Relative reproductive success.
- Common ancestry: Shared origin of lineages through descent.
- Homology: Similarity due to shared ancestry.

EVOLUTION
Change in heritable traits of populations across generations.

NATURAL SELECTION
Nonrandom differences in survival or reproduction tied to heritable traits.

ADAPTATION
Heritable trait shaped by selection in a context.

FITNESS
Relative reproductive success.

COMMON ANCESTRY
Shared origin of lineages through descent.

HOMOLOGY
Similarity due to shared ancestry.

DARWIN AND EVOLUTION

Lab connection

- Lab file: lab-12_darwin-evolution.md
- Lab evidence should test or illustrate: Natural selection mechanism
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

QUESTION
Historical explanation for adaptation

EVIDENCE
lab-12_darwin-evolution.md

CLAIM
Explain natural selection using variation, heritability, and reproductive success.

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Contrast check

- Do not stop at: Darwinian evolution explains adaptation through descent with modification.
- Stronger explanation: Evolution changes populations over generations, not individual organisms during life.
- The goal is to move from a named idea to a testable biological explanation.

SURFACE
Darwinian evolution explains adaptation through descent with modification.

DEEPER
Evolution changes populations over generations, not individual organisms during life.

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Module 12: Darwin and Evolution Retrieval and Lab Check

- Module focus: Darwin and Evolution
- Retrieval prompt: What are the required ingredients for natural selection?
- Answer check: Explain natural selection using variation, heritability, and reproductive success.
- Required terms: Evolution, Natural selection, Adaptation
- Lab connection: lab-12_darwin-evolution.md; lab-12_darwin-evolution.md supplies evidence for trait variation, environmental filters, and selection evidence.

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Practice quiz bridge

- 1. Natural selection requires all of the following EXCEPT:
- 2. In evolutionary terms, an organism's "fitness" means its:
- 3. The forelimb bones of humans, whales, and bats share the same underlying arrangement. This is evidence of:
- 4. "An individual giraffe stretched its neck during its life and so it evolved a longer neck." Why is this wrong?

QUESTION 1

Answer first

Natural selection requires all of the following EXCEPT:

QUESTION 2

Answer first

In evolutionary terms, an organism's "fitness" means its:

QUESTION 3

Answer first

The forelimb bones of humans, whales, and bats share the same underlying arrangement. This is evidence of:

QUESTION 4

Answer first

"An individual giraffe stretched its neck during its life and so it evolved a longer neck." Why is this wrong?

DARWIN AND EVOLUTION

Synthesis and exit ticket

- Exit claim: explain historical explanation for adaptation using evidence from lab-12_darwin-evolution.md.
- Must include: Evolution, Natural selection, Adaptation.
- Revision prompt: what evidence would change your explanation?

CLAIM
Historical explanation for adaptation

EVIDENCE
lab-12_darwin-evolution.md

REVISION
What would change your mind?